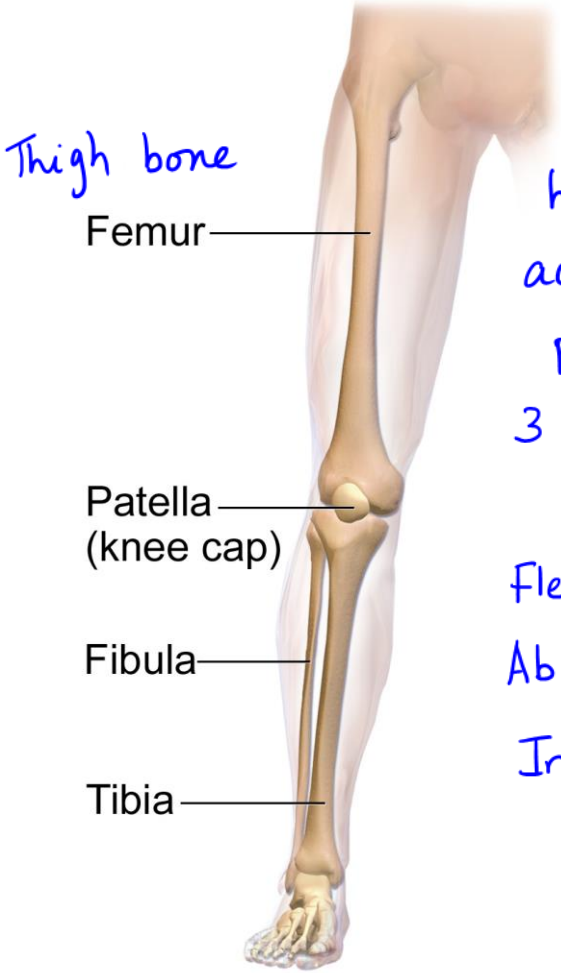
Anatomy – lower extremities



Leg Bones

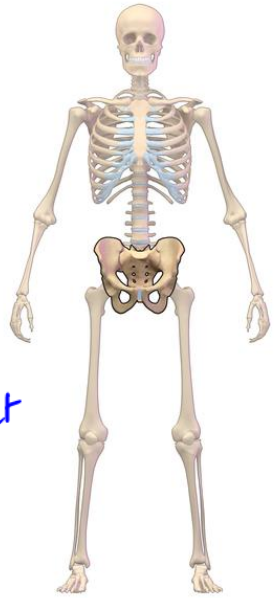
Hip joint - joint between femoral head and the acetabulum of the pelvis

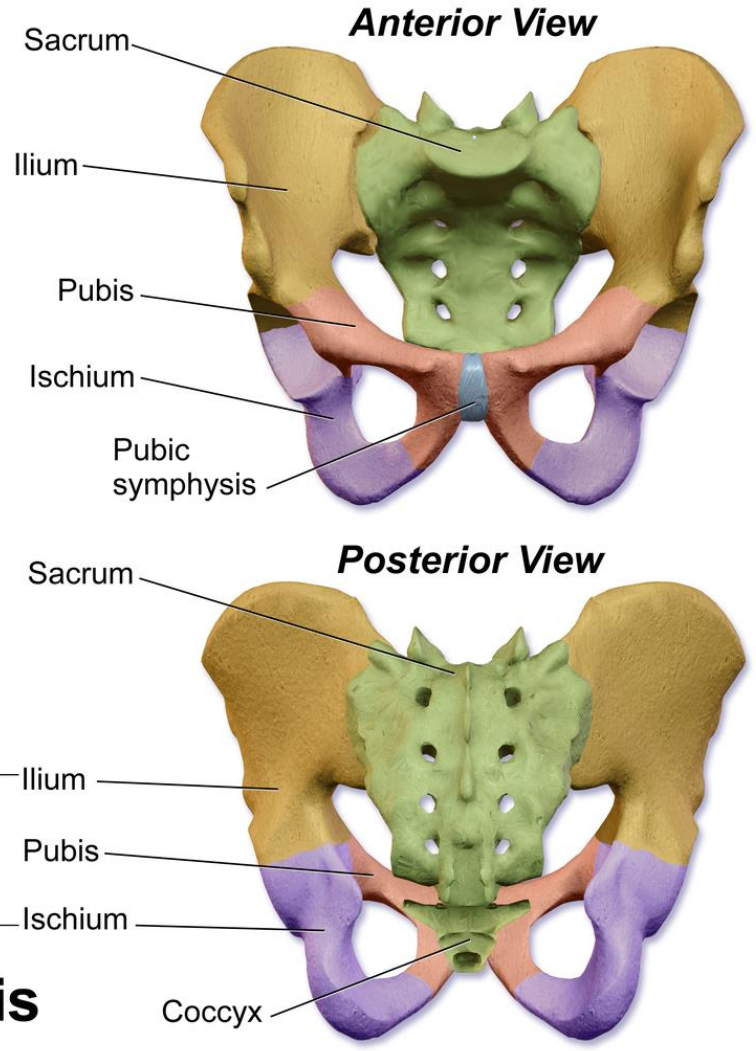
3 dof ball and socket jt

Flexion - extension

Abduction - adduction

Internal - external rotation

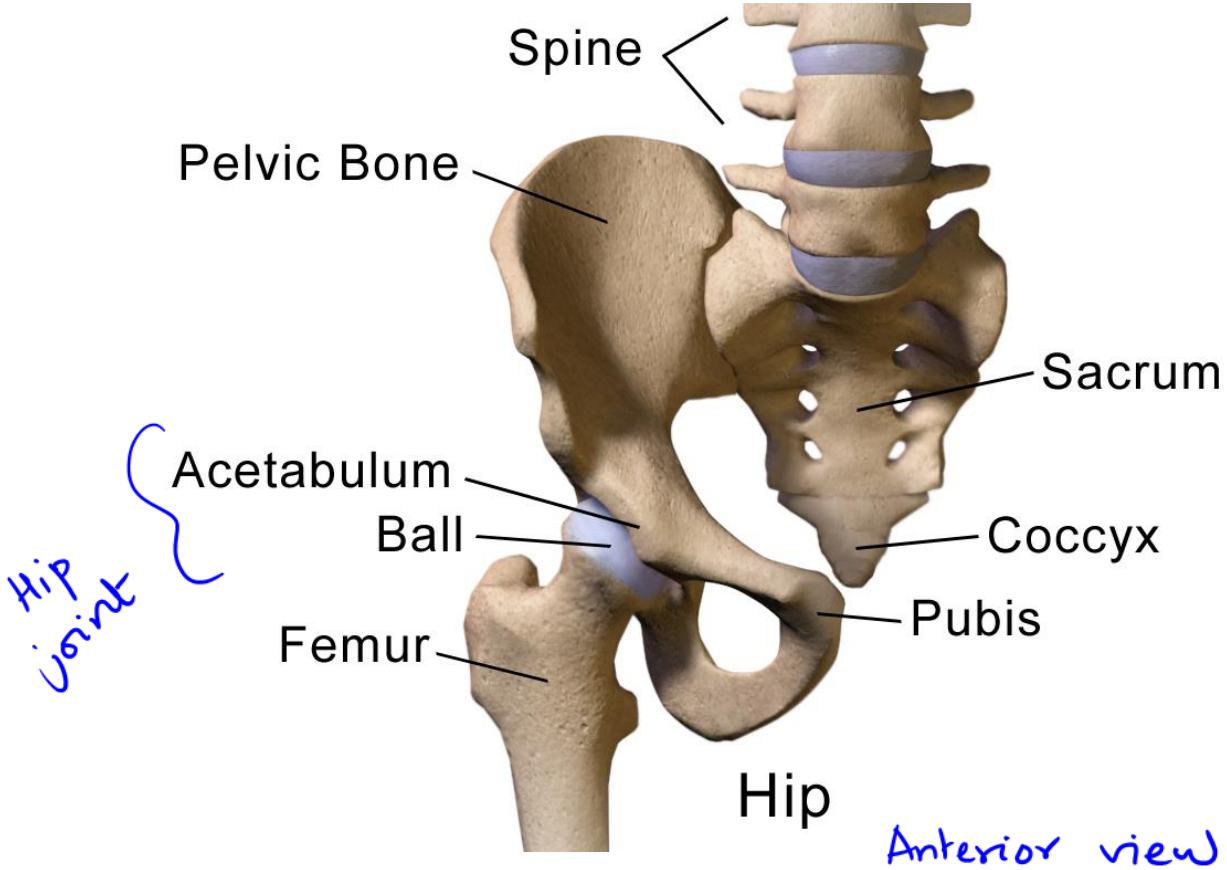




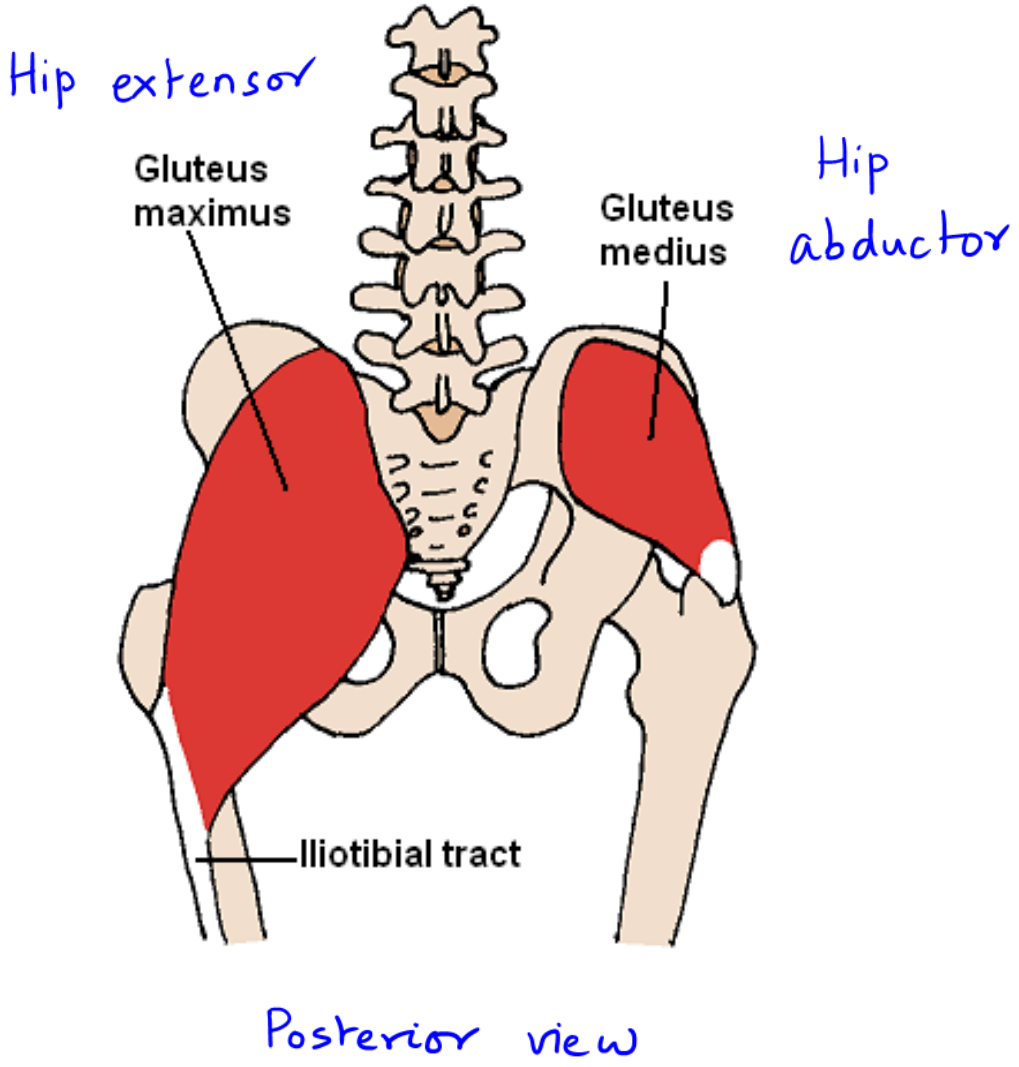
The Pelvis

Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2). DOI:10.15347/wjm/2014.010. ISSN 2002-4436. (Own work) [CC BY 3.0 (http://creativecommons.org/licenses/by/3.0)], via Wikimedia Commons

Anatomy – hip and pelvis

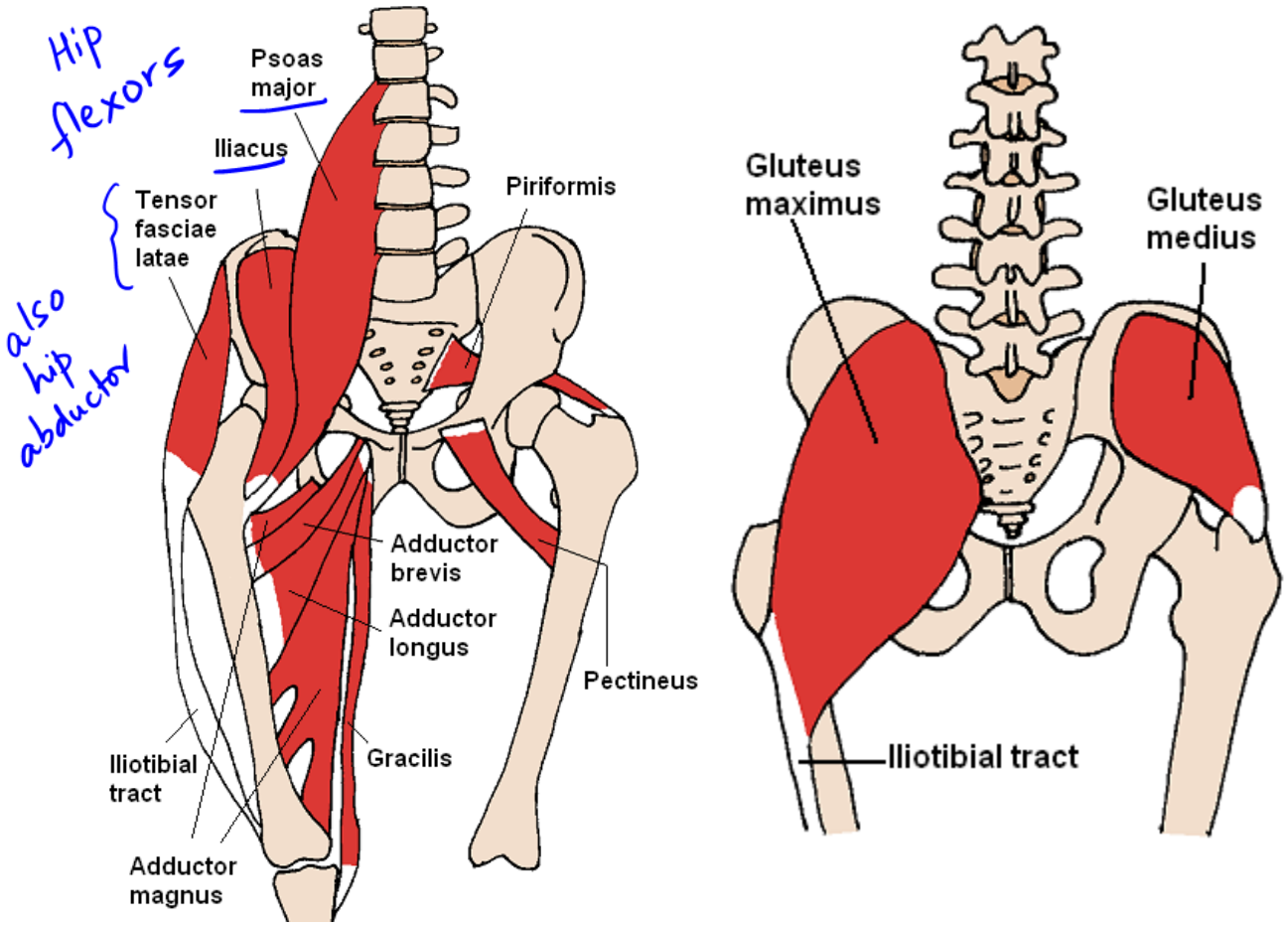


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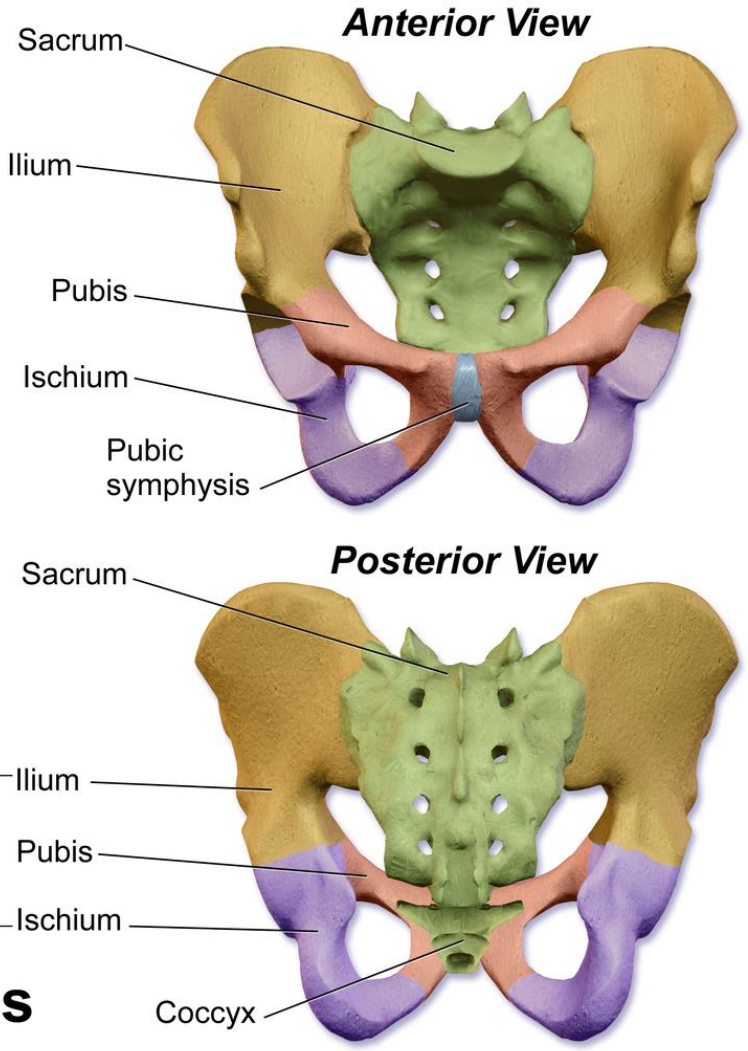
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Anatomy – hip and pelvis



By Beth ohara - Own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=545389

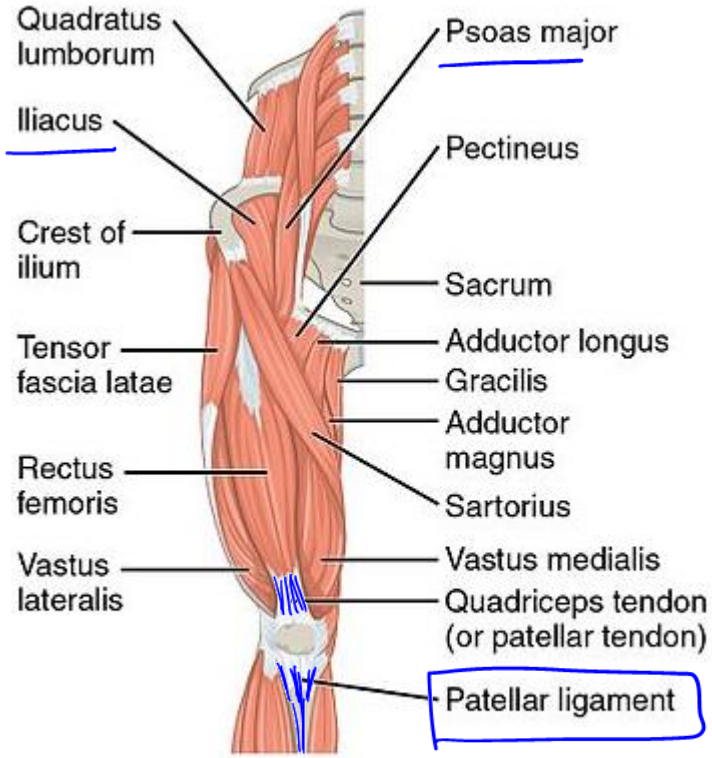




Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2). DOI:10.15347/wjm/2014.010. ISSN 2002-4436. (Own work) [CC BY 3.0 (http://creativecommons.org/licenses/by/3.0)], via Wikimedia Commons

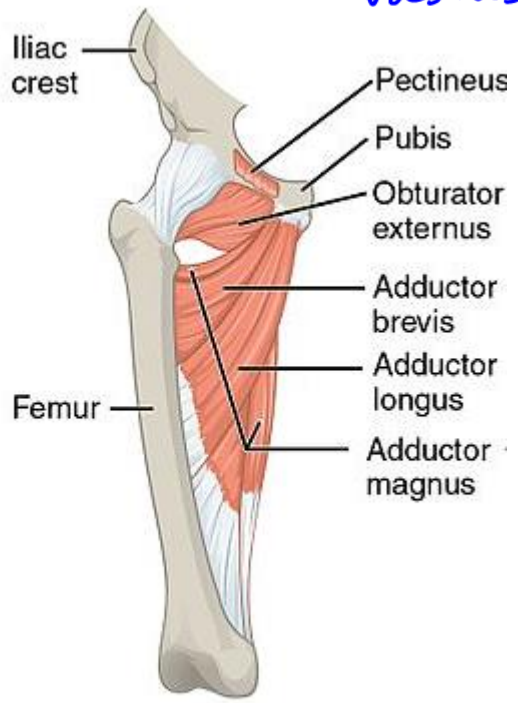
Anatomy – hip and pelvis

Quadriceps – rectus femoris
vastus medialis
vastus intermedius
vastus lateralis
knee extensors



Superficial pelvic and thigh muscles of right leg (anterior view)

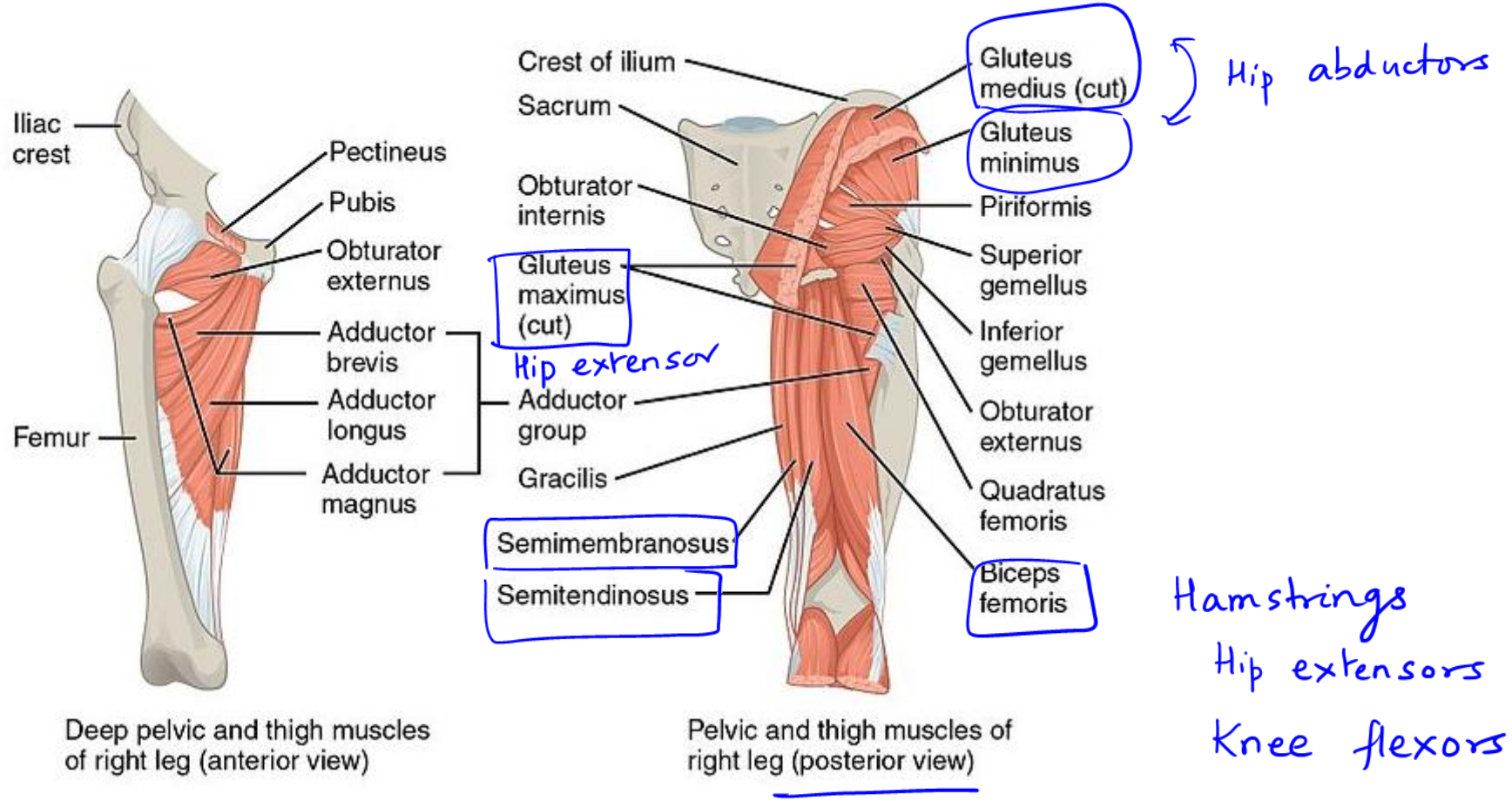
patella
knee cap
Sesamoid bone



Deep pelvic and thigh muscles of right leg (anterior view)

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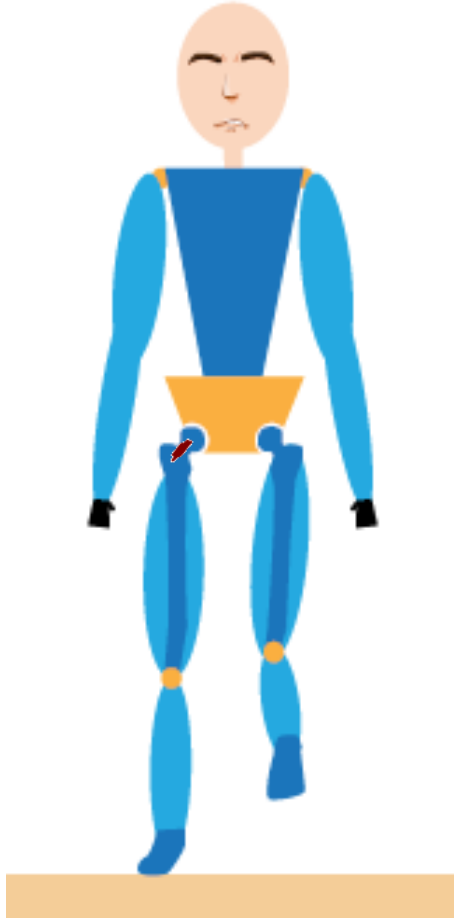
Anatomy – hip and pelvis

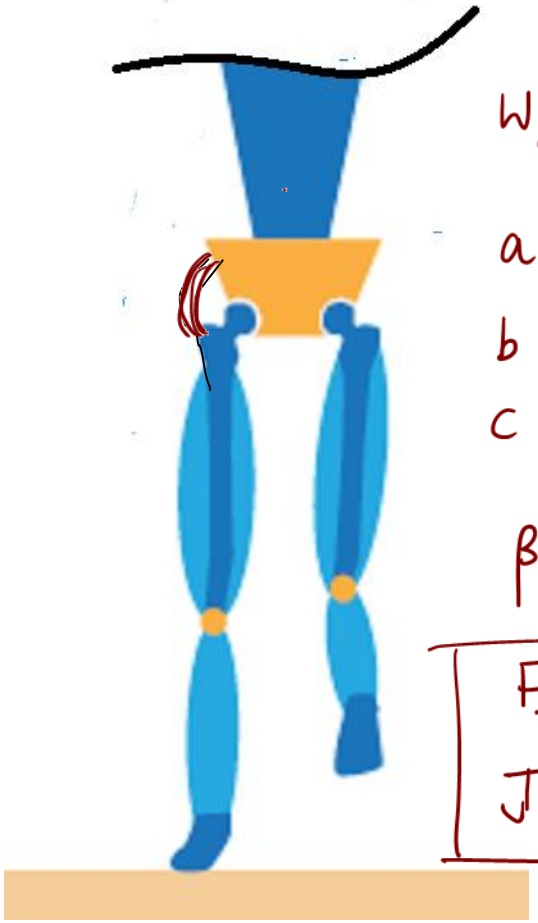


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Static analysis – hip in single stance

all of our body weight is on one leg





$W_{leg} = 0.16W$

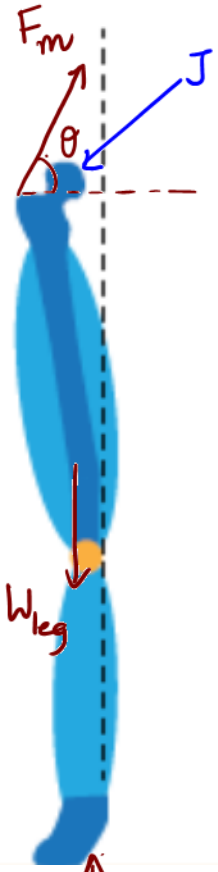
$a = 7$ cm

$b = 17.8$ cm

$c = 10.2$ cm

$\beta = 80^\circ$

$\theta = 70^\circ$





$$F_m = 1.6\,W$$

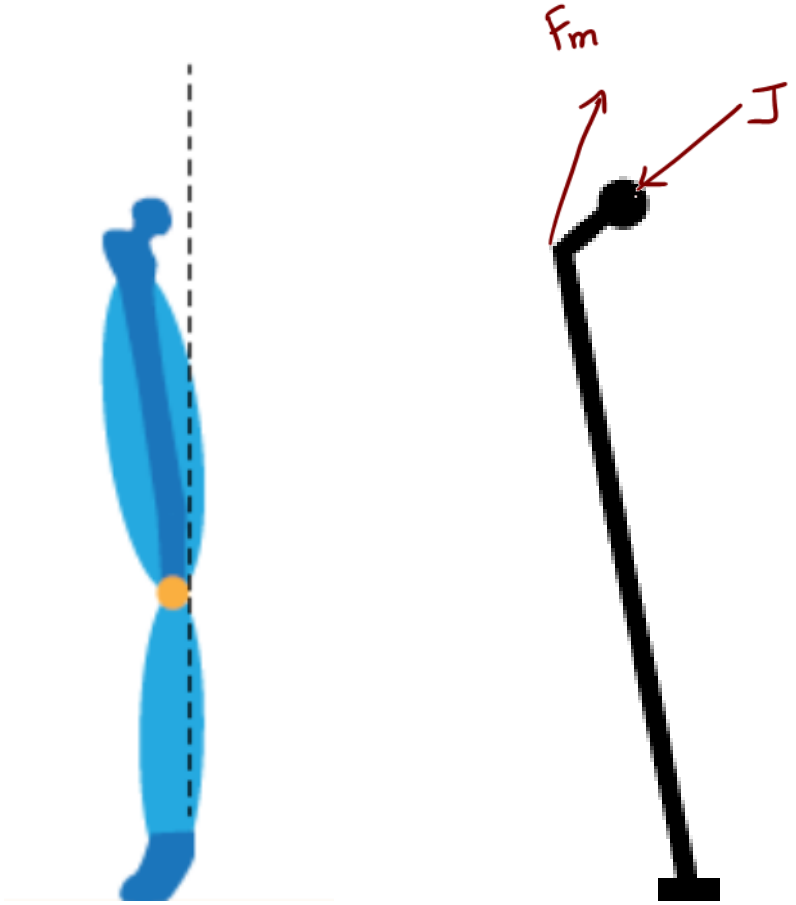
$$J = 2.4\,W$$

$\Sigma M_J = 0$

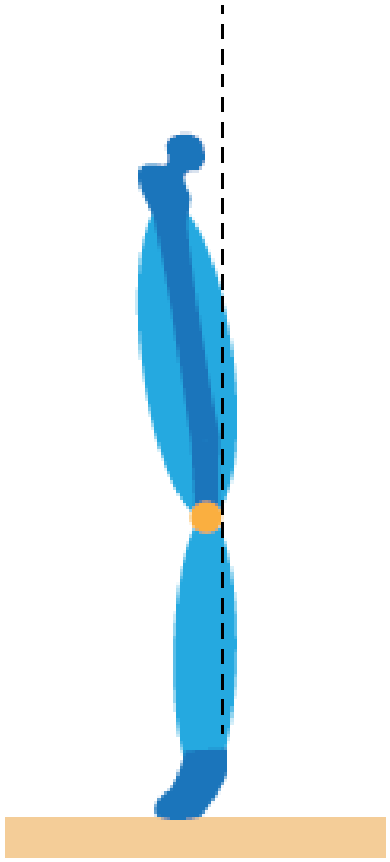
$\Sigma F_x = 0$

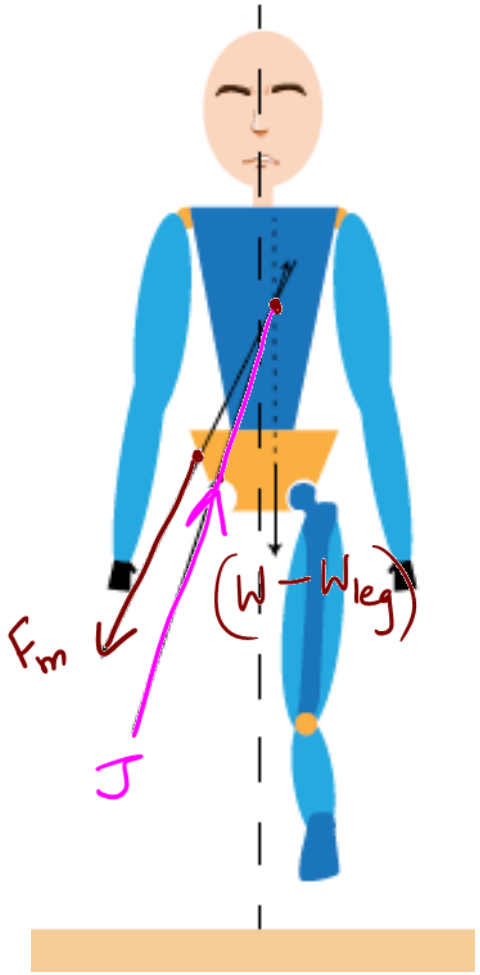
$\Sigma F_y = 0$

Static analysis – hip in single stance



Static analysis – hip in single stance





F_m, J & $(W - W_{leg})$ form a 3-force system

Saying the 3 forces are concurrent is equivalent to $\Sigma M = 0$

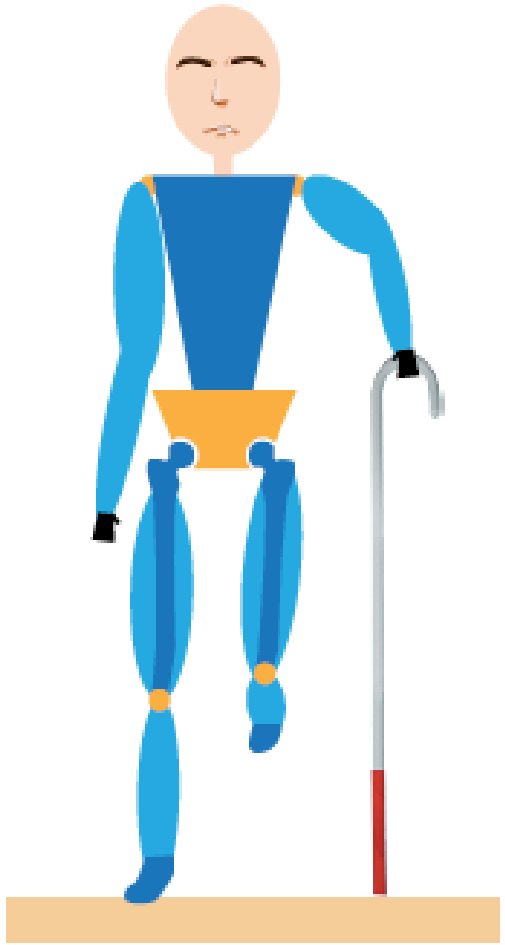


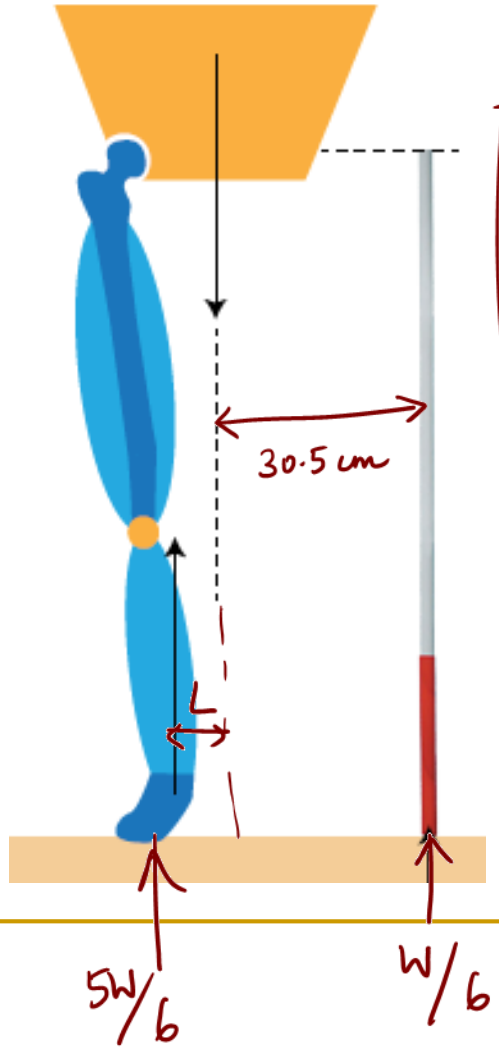
Static analysis – using a cane

$$\frac{5W}{6} L = \frac{W}{6}(30.5)$$

$\theta = 70^\circ$

$\therefore L = 6.1$ cm





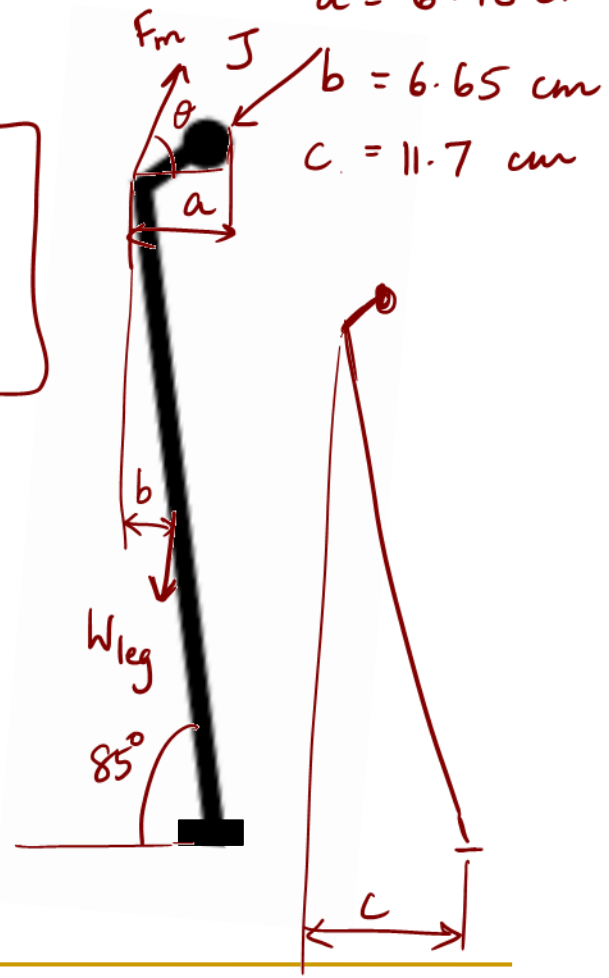
$F_m = 0.61\,W$

$J = 1.26\,W$

$a = 6.98$ cm

$b = 6.65$ cm

$c = 11.7$ cm



For hip abductor weakness or a painful hip, use the cane on the side opposite the injured side